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Learning to Walk from Three Minutes of Real-World Data with Semi-structured Dynamics Models

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Abstract: Traditionally, model-based reinforcement learning (MBRL) methods exploit neural networks as flexible function approximators to represent *a priori* unknown environment dynamics. However, training data are typically scarce in practice, and these black-box models often fail to generalize. Modeling architectures that leverage known physics can substantially reduce the complexity of system-identification, but break down in the face of complex phenomena such as contact. We introduce a novel framework for learning semi-structured dynamics models for contact-rich systems which seamlessly integrates structured first principles modeling techniques with black-box auto-regressive models. Specifically, we develop an ensemble of probabilistic models to estimate external forces, conditioned on historical observations and actions, and integrate these predictions using known Lagrangian dynamics. With this semi-structured approach, we can make accurate long-horizon predictions with substantially less data than prior methods. We leverage this capability and propose Semi-Structured Reinforcement Learning (`SSRL`) a simple model-based learning framework which pushes the sample complexity boundary for real-world learning. We validate our approach on a real-world Unitree Go1 quadruped robot, learning dynamic gaits – from scratch – on both hard and soft surfaces with just a few minutes of real-world data. Video and code are available at: https://sites.google.com/utexas.edu/ssrl

Keywords: Model-Based Reinforcement Learning, Physics-Based Models

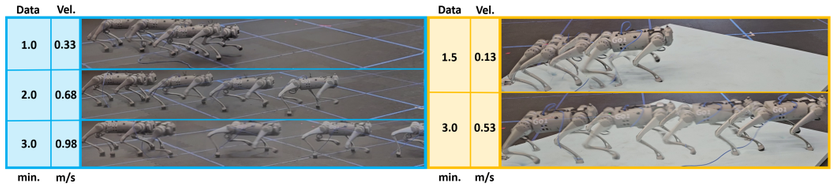


Figure 1: Unitree Go1 quadruped learning to walk from scratch using `SSRL` on hard ground (left) and memory foam (right).

1 Introduction

Effective robotic agents must leverage complex interactions between the robot and its environment, which are difficult to model using first principles. Model-based reinforcement learning (MBRL) is a powerful paradigm for controller synthesis [1], wherein the robot learns a generative dynamics model for the environment. The model can then be used to hallucinate synthetic rollouts [2], providing a source of data augmentation for policy optimization algorithms [3, 4, 5]. When the model is accurate, it can generate long rollouts which extrapolate beyond the training data, accelerating policy learning substantially. However, in practice, the black-box neural network models favored in the

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基于半结构化动力学模型、利用三分钟真实世界数据学习行走

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摘要：传统上，基于模型的强化学习（MBRL）方法利用神经网络作为灵活的函数逼近器，以表示先验未知的环境动力学。然而，在实践中训练数据通常稀缺，而这些黑箱模型往往难以泛化。利用已知物理知识的建模架构可以显著降低系统辨识的复杂度，但在面对接触等复杂现象时则会失效。我们提出了一种新颖的框架，用于为富接触系统学习半结构化动力学模型，该框架无缝集成了结构化第一原理建模技术与黑箱自回归模型。具体而言，我们开发了一组概率模型来估计外力，这些模型以历史观测和动作作为条件，并利用已知的拉格朗日动力学整合这些预测。通过这种半结构化方法，我们可以用比先前方法少得多的数据做出准确的长期预测。我们利用这一能力，提出了半结构化强化学习（`SSRL`），这是一种简单的基于模型的学习框架，推动了真实世界学习的样本复杂度边界。我们在真实的Unitree Go1 四足机器人上验证了我们的方法，仅使用几分钟的真实世界数据，就从零开始在硬质和软质表面上学习了动态步态。视频和代码可在以下网址获取：https://sites.google.com/utexas.edu/ssrl

关键词：基于模型的强化学习，基于物理的模型

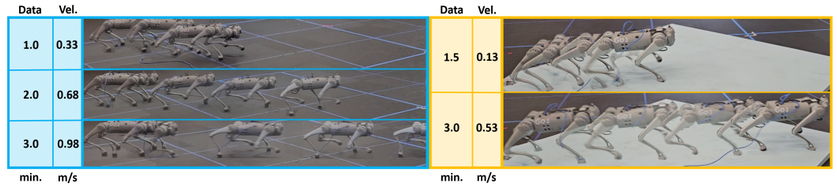


图 1：Unitree Go1 四足机器人使用 SSRL 在硬地面（左）和记忆海绵（右）上从零开始学习行走。

1 引言

有效的机器人智能体必须利用机器人与其环境之间复杂的相互作用，而这些相互作用难以通过第一性原理进行建模。基于模型的强化学习（MBRL）是一种用于控制器综合的强大范式 [1]，，其中机器人学习环境的一个生成式动力学模型。该模型随后可用于生成合成 rollout [2]，，为策略优化算法提供数据增强来源 [3, 4, 5]。当模型准确时，它能够生成长于训练数据的轨迹，从而显著加速策略学习。然而在实践中，黑箱神经网络模型在

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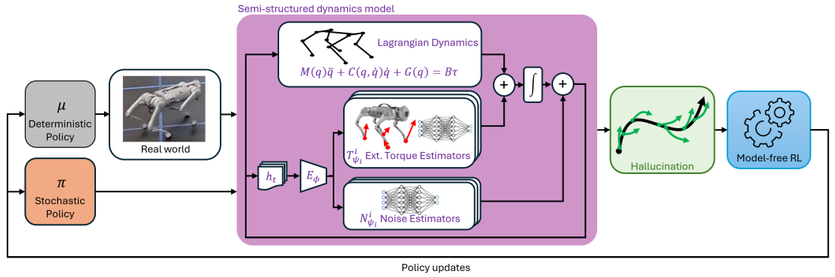


Figure 2: The SSRL framework. A deterministic policy is used to collect data from the real world while a stochastic policy is utilized in conjunction with the learned dynamics model to "hallucinate" short synthetic rollouts which branch from this data. The model incorporates Lagrangian dynamics and encodes previous state predictions, which are fed to external torque and noise estimators to predict future states. The synthetic data is used with a model-free RL algorithm to update the policies.

MBRL literature struggle to generalize beyond the training data [6, 7, 8], and thus do not outperform modern model-free alternatives [9, 10]. Currently, both paradigms are too inefficient and unreliable to make learning new behaviors in the real world practical for many applications.

An appealing alternative is to leverage known physics to design structured model classes. This general approach has been used for efficient system identification and controller synthesis across many bodies of work, ranging from classic adaptive control techniques [11, 12, 13, 14] to more recent physics-informed neural architectures [15, 16, 17]. However, these previous approaches do not scale to the complexities of real-world learning for contact rich-systems such as walking robots (Fig. 1). Indeed, modeling contact remains an open problem. Moreover, prior structured approaches to learning contact dynamics make strong assumptions about what can be perceived with available on-board sensors. For example, at inference time [18] requires access to a signed-distance representation of potential contact surfaces. More generally, prior approaches generally assume access to privileged state observations which make it possible to predict exactly when and where contact is made [19, 15]. However, reliably estimating these quantities in the real-world using noisy on-board sensors is another open area of research [20, 21, 22, 23, 24].

We ask: *can we build a light-weight modeling framework which leverages known structure and is implementable with onboard sensors?* We answer this question in the affirmative by introducing Semi-structured Reinforcement Learning (SSRL), a simple MBRL pipeline for contact-rich control in the real-world (Fig. 2). For concreteness, we focus primarily on the quadruped depicted in Fig. 1 and aim to learn an effective locomotion controller **entirely from scratch in the real world**. We consider the case where the robot's observations only include proprioceptive measurements via joint encoders, IMU measurements, and a global velocity estimator.

We inspire our approach by looking to the Lagrangian equation of motion for the robot:

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = B\tau + J^T F^e + \tau^d, \tag{1}$$

where q and $\dot{q}$ are the generalized coordinates and velocities of the robot, motor torques τ are distributed to the joints by the matrix B, F^e represents contact forces generated by the environment with J the resulting Jacobian, τ^d represents parasitic damping torques, and M, C, and G are the mass matrix, Coriolis and centrifugal forces, and gravity vector. Here, B, M, C, and G are determined by the geometry and inertial properties of the robot, which are known *a priori*. Thus, given access to observations, we assume that the only unknown terms are $J^T F^e$ and τ^d.

We leverage the known Lagrangian structure of (1) by estimating these unknown terms with an estimator $\hat{\tau}^e(h_t) \approx J^T F^e + \tau^d$ which is conditioned on a history h_t of the available measurements, enabling the model to infer information about hidden states of the environment such as the geometry of the ground underfoot [25, 26]. We instantiate $\hat{\tau}^e$ as an ensemble of probabilistic models to help

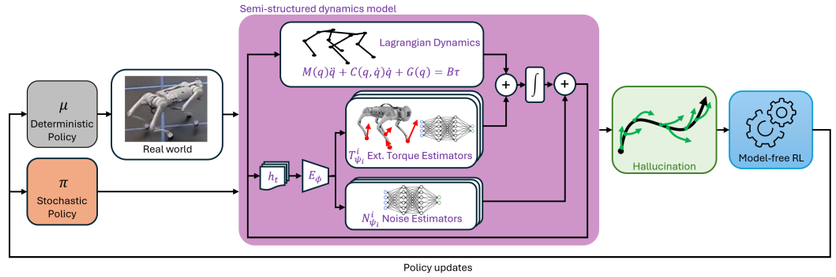


图 2：SSRL 框架。使用确定性策略从真实世界收集数据，同时结合学习到的动力学模型利用随机策略“虚构”出从该数据分支出来的短合成 rollout。该模型融合了拉格朗日动力学，并对先前的状态预测进行编码，这些预测被送入外部力矩和噪声估计器以预测未来状态。合成数据与无模型强化学习算法一起用于更新策略。

MBRL 文献中难以泛化到训练数据之外 [6, 7, 8], ，因此无法超越现代无模型替代方法 [9, 10]。目前，这两种范式都效率低下且不可靠，使得在许多应用中于真实世界学习新行为尚不切实际。

一种有吸引力的替代方案是利用已知物理规律来设计结构化模型类别。这种通用方法已被用于众多研究中的高效系统辨识和控制器综合，范围涵盖从经典自适应控制技术 [11, 12, 13, 14] 到更近期的物理信息神经网络架构 [15, 16, 17]。然而，这些先前的方法无法扩展到丰富接触系统（如行走机器人）的真实世界学习的复杂性中（图 1）。事实上，对接触进行建模仍然是一个未解难题。此外，先前用于学习接触动力学的结构化方法对机载传感器所能感知的内容做出了强烈假设。例如，在推理时 [18]需要访问潜在接触表面的符号距离表示。更一般地说，先前的方法通常假设可以访问特权状态观测，从而能够精确预测接触发生的时间和位置 [19, 15]。然而，在真实世界中利用嘈杂的机载传感器可靠地估计这些量是另一个开放的研究领域 [20, 21, 22, 23, 24]。

我们提出一个问题：能否构建一个轻量级的建模框架，该框架利用已知结构并可通过机载传感器实现？我们通过引入半结构化强化学习（SSRL）对此问题给出肯定回答，这是一种用于真实世界中丰富接触控制的简单基于模型的强化学习流程（图 2）。为具体起见，我们主要聚焦于图 1 所示的四足机器人，旨在**完全从零开始在真实世界中**学习一种有效的运动控制器。我们考虑的情况是，机器人的观测仅包含通过关节编码器、IMU 测量和全局速度估计器获得的本体感觉测量。

我们的方法灵感来源于机器人的拉格朗日运动方程：

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = B\tau + J^T F^e + \tau^d, \tag{1}$$

其中 q 和 $\dot{q}$ 是机器人的广义坐标和速度，电机扭矩 τ 通过矩阵 B 分配至各关节， F^e 表示由环境产生的接触力， J 是相应的雅可比矩阵， τ^d 表示寄生阻尼扭矩，而 M、 C 和 G 分别是质量矩阵、科里奥利力和离心力以及重力向量。此处， B， M， C 和 G 由机器人的几何和惯性特性决定，这些特性是已知的 先验 知识。因此，在能够获取观测数据的前提下，我们假设唯一未知的项是 $J^T F^e$ 和 τ^d。

我们利用方程 (1) 中已知的拉格朗日结构，通过一个估计器来估算这些未知项，该估计器 $\hat{\tau}^e(h_t) \approx J^T F^e + \tau^d$ 以可用测量的历史 h_t 为条件，使模型能够推断环境的隐状态信息，例如脚下地面的几何形状 [25, 26]。我们将 $\hat{\tau}^e$ 实例化为概率模型的集成，以帮助

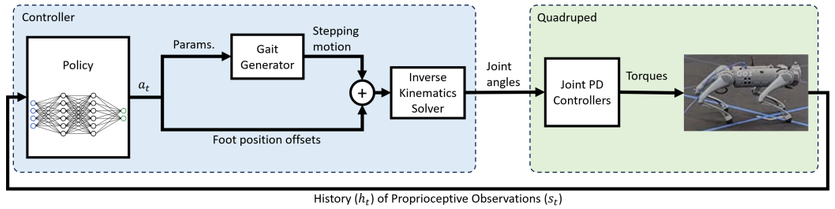


Figure 3: Control architecture. The policy takes in a history of observations and outputs parameters to a gait generator and offsets to the gait. The resulting foot positions are sent to an inverse kinematics solver which computes desired joint angles for joint level PD controllers.

capture uncertainty in the predictions, as is commonplace in the MBRL literature [6, 27]. We then introduce a novel inference procedure for making forward predictions, wherein the predictions made by these semi-structured models are fed back into $\hat{\tau}^e$ in an auto-regressive fashion. The SSRL framework leverages these predictions as a source of data augmentation to accelerate real-world policy optimization, using a simple Dyna-style [3, 6] MBRL algorithm. Finally, we demonstrate that this light-weight approach scales to the real-world, learning to walk on different terrains within a minute and learning substantially more dynamic gates within a few minutes. This result outperforms recent state-of-the-art real-world learning results for quadrupeds [28] by both a) requiring an order-of-magnitude less real-world data and b) reaching substantially higher walking speeds.

2 Preliminaries and Problem Formulation

Notation: While the underlying dynamics of the robot evolve in continuous time according to the differential equation (1), the policy acts in discrete time and we will use subscripts to denote discrete time steps. The state of the robot $s_t \in \mathcal{S}$ captures the available proprioceptive states q_t and their velocities $\dot{q}_t$, available from joint encoders and IMU measurements. This does not include global coordinates of the robot. We capture the state of the environment (or *extrinsics*) $e_t \in \mathcal{E}$, which includes non-proprioceptive information (such as the distance of the ground below the robot) and the state of the ground underfoot (such as terrain deformation). The robot actions $a_t \in \mathcal{A}$ are the outputs of neural network described below. The overall dynamics for the system are defined by:

$$\text{Robot Transitions: } s_{t+1} \sim p_s(\cdot|s_t, a_t, e_t) \qquad \text{Environment Transitions: } e_{t+1} \sim p_e(\cdot|s_t, a_t, e_t). \quad (2)$$

To control this joint system, we will denote the policy the agent optimizes via $a_t \sim \pi_\theta(\cdot|s_t, h_t)$, where $h_t = (s_{t-1}, \ldots, s_{t-h})$ bundles the histories of state measurements and θ are parameters.

Control Architecture: We adopt the control architecture depicted in Fig. 3, similar to prior works on RL-based locomotion [25, 29, 30]. We optimize a neural network policy $\pi_\theta(\cdot|s_t, h_t)$ conditioned on previous observations which outputs (i) changes to parameters of a nominal gait generator that outputs desired foot positions and (ii) offsets to these foot positions. The resulting targets are sent to an inverse kinematics solver which computes desired joint positions. The joint targets are output at 100 Hz to low-level PD controllers for conversion to torques. Additional details are provided in Appendix A.

Reinforcement Learning Problem: We formally frame the learning of a locomotion controller in the real world in terms of a partially observable Markov decision process (POMDP) [31], defined by the tuple $(\mathcal{X}, \mathcal{A}, p, r, \Omega, O, \gamma)$. Here, $\mathcal{X} = \mathcal{S} \times \mathcal{E}$ is the overall state space for the system, and $p(\cdot|s_t, a_t, e_t) = (p_s(\cdot|s_t, a_t, e_t), p_e(\cdot|s_t, a_t, e_t))$ captures the joint robot-environment dynamics. The space of observations Ω consists of the states that can be measured, and the observation distribution $O(\cdot|s_t, a_t, e_t)$ provides (noisy) estimates of the states from onboard sensors. The reward function $r(s_t, a_t, s_{t+1})$ depends only upon the robot states and actions, and is therefore directly measurable in the real world. We define the reward function to maximize the robot's forward velocity, maintain upright orientation, minimize angular rates, conserve energy, and avoid excessive torques.

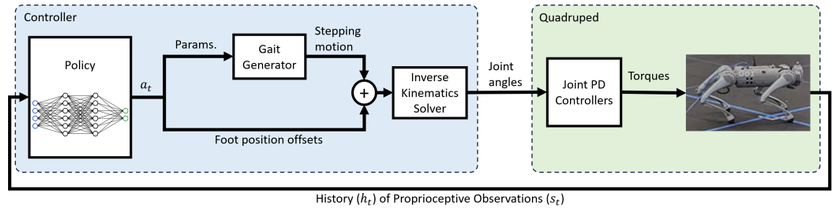


图 3：控制架构。策略接收观测历史，并输出步态生成器的参数及步态偏移量。生成的足端位置被发送至逆运动学求解器，该求解器计算期望关节角度以供关节级 PD 控制器使用。

捕捉预测中的不确定性，这在基于模型的强化学习文献中很常见 [6, 27]。接着，我们引入一种新颖的前向预测推理过程，其中这些半结构化模型做出的预测会以自回归的方式反馈到$\hat{\tau}^e$ 中。SSRL 框架利用这些预测作为数据增强的来源，以加速真实世界的策略优化，并使用一种简单的 Dyna 风格 [3, 6] 基于模型的强化学习算法。最后，我们证明这种轻量级方法可扩展至真实世界，能够在一分钟内学会在不同地形上行走，并在几分钟内学会显著更具动态性的步态。这一结果超越了近期四足机器人的最先进真实世界学习成果 [28]，体现在两方面： a）所需的真实世界数据减少了一个数量级，以及 b）实现了显著更高的行走速度。

2 预备知识与问题表述

符号说明：尽管机器人的底层动力学根据微分方程 (1) 在连续时间中演化，但策略在离散时间下运行，我们将使用下标表示离散时间步骤。机器人状态 $s_t \in \mathcal{S}$ 包含可用的本体感受状态 q_t 及其速度 $\dot{q}_t$，这些数据来自关节编码器和 IMU 测量。这不包括机器人的全局坐标。我们捕捉环境状态（或外参） $e_t \in \mathcal{E}$，其中包括非本体感受信息（例如机器人下方地面的距离）以及脚下地面的状态（例如地形形变）。机器人动作 $a_t \in \mathcal{A}$是下述神经网络的输出。系统的整体动力学定义如下：

$$\text{Robot Transitions: } s_{t+1} \sim p_s(\cdot|s_t, a_t, e_t) \qquad \text{Environment Transitions: } e_{t+1} \sim p_e(\cdot|s_t, a_t, e_t). \quad (2)$$

为了控制该关节系统，我们将代理优化的策略表示为 $a_t \sim \pi_\theta(\cdot|s_t, h_t)$，其中 $h_t = (s_{t-1}, \ldots, s_{t-h})$ 捆绑了状态测量的历史，而 θ 是参数。

控制架构：我们采用如图 3所示的控制架构，类似于先前关于基于强化学习的运动控制的研究 [25, 29, 30]。我们优化一个神经网络策略 $\pi_\theta(\cdot|s_t, h_t)$，该策略以先前的观测为条件，输出（i）标称步态生成器参数的变化量（该生成器输出期望的足端位置），以及（ii）对这些足端位置的偏移量。生成的目标被发送至逆运动学求解器，以计算期望的关节位置。关节目标以100Hz 的频率输出至底层 PD 控制器，用于转换为扭矩。更多细节见附录 A。

强化学习问题：我们将真实世界中运动控制器的学习正式表述为一个部分可观测马尔可夫决策过程（POMDP） [31]，，该过程由元组 $(\mathcal{X}, \mathcal{A}, p, r, \Omega, O, \gamma)$ 定义。其中， $\mathcal{X} = \mathcal{S} \times \mathcal{E}$ 是系统的整体状态空间， $p(\cdot|s_t, a_t, e_t) = (p_s(\cdot|s_t, a_t, e_t), p_e(\cdot|s_t, a_t, e_t))$ 描述了机器人与环境的联合动力学。观测空间 Ω 由可测量的状态组成，而观测分布 $O(\cdot|s_t, a_t, e_t)$ 则提供来自机载传感器的状态（含噪声）估计。奖励函数 $r(s_t, a_t, s_{t+1})$ 仅取决于机器人状态和动作，因此在真实世界中可直接测量。我们定义奖励函数以最大化机器人的前向速度、保持直立姿态、最小化角速度、节约能量并避免过大扭矩。

Finally, we define a termination flag $d_t \in \{0, 1\}$ where $d_t = 1$ when body roll or pitch exceed limits. Exact definitions are found in Appendix A. Given an episode length $T \in \mathbb{N}$, discount factor $\gamma \in (0, 1)$, and a distribution x_0 over $\mathcal{X}$ of initial conditions for the system, the goal is to maximize the expected discounted total reward: $\max_\theta \mathbb{E}[\sum_{t=0}^{T} \gamma^t (1 - d_t) \cdot r(s_t, a_t, s_{t+1})]$.

3 Semi-structured Reinforcement Learning

A high-level overview of our method is presented in Fig. 2, outlined as follows: i) we learn an ensemble of deterministic external torque estimators $\hat{\tau}^e$ which are conditioned on h_t; ii) we then integrate the torque predictions through the Lagrangian ODE (1), add a learned noise term to generate probabilistic 1-step predictions, and feed this prediction back into h_t to produce auto-regressive predictions over multiple steps iii) we fit this semi-structured representation of the dynamics using multi-step prediction losses; and finally iv) predictions from these models are used as a source of data augmentation for MBRL.

3.1 External Torque Estimators

We construct our approximations to the discrete-time probabilistic robot transition dynamics $s_{t+1} \sim p_s(\cdot|s_t, a_t, e_t)$ by building on top of the deterministic Lagrangian dynamics (1) which we rewrite as:

$$M(q)\ddot{q} + C(q, \dot{q}) + G(q) = B\tau + \underbrace{J(s, e)^T F^e(s, e, \tau) + \tau^d(s, e)}_{\tau^e(s,e,\tau)}, \tag{3}$$

where we now explicitly denote the dependence of the contact Jacobian $J(s, e)$, contact forces $F^e(s, e, \tau)$, and dissipative terms $\tau^d(s, e)$ on the state of the robot, environment, and low-level torques supplied by the motor. As discussed in Section 1, identifying precise locations where contact occurs on the robot can be extremely difficult from on-board measurements. Thus, instead of inferring J and F^e separately, we directly estimate $\tau^e = J^T F^e + \tau^d$. We condition these estimates a latent encoding $z_t = E_\phi(h_t)$ of h_t, which enables the network to infer information about the extrinsics e_t needed for predicting future states [26, 25]. Altogether, we use an ensemble of deterministic models $\tau^e \approx \bar{\tau}_t^{e,i} = T_{\psi_i}(s_t, a_t, z_t)$ with tunable parameters ψ_i.

3.2 Generating Forwards Predictions

We first describe how we generate a probabilistic 1-step prediction for the next state associated to each of the torques estimates $\tau^e \approx \bar{\tau}_t^{e,i} = T_{\psi_i}(s_t, a_t, z_t)$. First, we let $\tau_t = G(s_t, a_t)$ denote a zero-order hold estimate for the low-level motor torques applied to the robot; here, G is a known map which captures how the current state and action are processed by our control architecture (Fig. 3) to produce low-level joint torques. The i-th deterministic next-state prediction is then given by $\bar{s}_{t+1}^i = I(s_t, \tau_t, \bar{\tau}_t^{e,i})$, which captures how a chosen numerical integrator for the Lagrangian dynamics (1) propagates s_t, τ_t, and $\bar{\tau}_t^{e,i}$. We then add zero-mean Gaussian noise to $\bar{s}_{t+1}^i$ whose variance $\Sigma_t^i = N_{\psi_i}^i(s_t, a_t, z_t)$ is a diagonal Gaussian which is output by the same network predicting the i-th torque estimate. Altogether, this constructs a probabilistic estimate for the next state:

$$\text{Uncertainty-Aware State Predictions:} \qquad \hat{s}_{t+1}^i \sim \hat{p}_{\psi_i}^i(\cdot|s_t, a_t, h_t) := \mathcal{N}(\bar{s}_{t+1}^i, \Sigma_t^i) \tag{4}$$

The ensemble of probabilistic models $\{\hat{p}_{\psi_i}^i\}$ is used to generate k-step predictions using Algorithm 1, which adapts the method from [6] to our auto-regressive setting where future state predictions are conditioned on the history h_t. Given a state s_t and state history h_t, to generate a synthetic rollout we: (i) sample an action from the policy, (ii) randomly choose a model $\hat{p}_{\psi_i}^i$ from the ensemble, (iii) generate a next-state prediction $\hat{s}_{t+1}^i$ using (4), and iv) incorporate this prediction into h_{t+1} and repeat the previous steps until a k-step prediction has been generated.

3.3 Approximate Maximum Likelihood Estimation

To train $\{\hat{p}_\psi^i\}_i$, we maximize the joint-likelihood of state predictions along these synthetic rollouts under the real-world data. Due to the Markov assumption (Section 2), the joint distribution of

最后，我们定义一个终止标志 $d_t \in \{0, 1\}$，其中当 $d_t = 1$ 机体横滚或俯仰角超出限制时触发终止。确切定义见附录 A。给定回合长度 $T \in \mathbb{N}$、折扣因子$\gamma \in (0, 1)$以及系统初始条件在 x_0 上的分布 $\mathcal{X}$，目标是最大化期望折扣总奖励：$\max_\theta \mathbb{E}[\sum_{t=0}^{T} \gamma^t (1 - d_t) \cdot r(s_t, a_t, s_{t+1})]$。

3 半结构化强化学习

我们方法的高层概述见图 2，概括如下：i) 我们学习一个确定性外部力矩估计器集成 $\hat{}\tau^e$，其以 h_t为条件；ii) 随后，我们将力矩预测通过拉格朗日常微分方程（1）进行积分，添加一个学习得到的噪声项以生成概率单步预测，并将该预测反馈至 h_t，从而产生跨多步的自回归预测；iii) 我们利用多步预测损失拟合这种动力学的半结构化表示；最后，iv) 这些模型的预测被用作基于模型的强化学习的数据增强来源。

3.1 外部力矩估计器

我们通过构建在确定性拉格朗日动力学（1）之上来构造对离散时间概率机器人转移动力学 $s_{t+1} \sim p_s(\cdot|s_t, a_t, e_t)$ 的近似，并将其重写为：

$$M(q)\ddot{q} + C(q, \dot{q}) + G(q) = B\tau + \underbrace{J(s, e)^T F^e(s, e, \tau) + \tau^d(s, e)}_{\tau^e(s,e,\tau)}, \tag{3}$$

其中，我们现在明确标示接触雅可比矩阵 $J(s, e)$、接触力 $F^e(s, e, \tau)$ 以及耗散项 $\tau^d(s, e)$ 对机器人状态、环境状态以及电机提供的底层力矩的依赖关系。正如第 1 节所讨论，从机载测量中识别机器人上发生接触的确切位置可能极其困难。因此，我们不分别推断 J 和 F^e，而是直接估计 $\tau^e = J^T F^e + \tau^d$。我们将这些估计以 h_t 的隐编码 $z_t = E_\phi(h_t)$ 为条件，这使得网络能够推断预测未来状态 [26, 25] 所需的外参 e_t 信息。总体而言，我们使用一个确定性模型集成 $\tau^e \approx \bar{\tau}_t^{e\,i} = T_{\psi_i}(s_t, a_t, z_t)$，其具有可调参数 ψ_i。

3.2 生成前向预测

我们首先描述如何为与每个扭矩估计 $\tau^e \approx \bar{\tau}_t^{e\,i} = T_{\psi_i}(s_t, a_t, z_t)$，相关联的下一状态生成概率性单步预测。首先，令 $\tau_t = G(s_t, a_t)$ 表示应用于机器人的底层电机力矩的零阶保持估计；此处，G是一个已知映射，用于捕捉当前状态和动作如何通过我们的控制架构（图 3）处理以产生底层关节扭矩。第 i个确定性下一状态预测由 $\bar{s}_{t+1}^i = I(s_t, \tau_t, \bar{\tau}_t^{e,i})$给出，它描述了拉格朗日动力学（1）所选数值积分器如何传播 s_t、τ_t，和 $\bar{\tau}_t^{e\,i}$。随后，我们将零均值高斯噪声添加到 $\bar{s}_{t+1}^i$中，其方差 $\Sigma_t^i = N_{\psi_i}^i(s_t, a_t, z_t)$为一个对角高斯分布，由预测第 i个扭矩估计的同一网络输出。综上所述，这构建了对下一状态的概率估计：

$$\text{Uncertainty-Aware State Predictions:} \qquad \hat{s}_{t+1}^i \sim \hat{p}_{\psi_i}^i(\cdot|s_t, a_t, h_t) := \mathcal{N}(\bar{s}_{t+1}^i, \Sigma_t^i) \tag{4}$$

概率模型集成 $\{\hat{p}_{\psi_i}^i\}$ 用于生成 k步预测，使用算法 1，该算法将 [6] 中的方法适配到我们的自回归设置，其中未来状态预测以状态历史 h_t为条件。给定一个状态 s_t 和状态历史 h_t，要生成合成 rollout，我们：(i) 从策略中采样一个动作，(ii) 从集成中随机选择一个模型 $\hat{p}_{\psi_i}^i$，(iii) 使用(4) 生成下一状态预测 $\hat{s}_{t+1}^i$，以及 (iv) 将此预测纳入 h_{t+1}并重复上述步骤，直到生成了 k步预测。

3.3 近似最大似然估计

为了训练 $\{\hat{p}_\psi^i\}_i$，我们在真实世界数据下最大化这些合成轨迹中状态预测的联合似然。由于马尔可夫假设（第 2 节），其联合分布为

Algorithm 1 Auto-Regressive State Predictions

Inputs hallucination buffer $\mathcal{D}_{\texttt{model}}$, models $\{\hat{p}^i_{\psi_i}\}$, policy π_θ, start state s_0, start history h_0
for $t = 0 \dots k-1$ **do**
  Sample action $a_t \sim \pi_\theta(\cdot \mid s_t, h_t)$
  Randomly choose model $i \sim \mathcal{U}[1, \dots, P]$ and predict next state $\hat{s}^i_{t+1}$ with (4)
  Update state history h_{t+1} with $\hat{s}^i_{t+1}$ and $s_{t+1} \leftarrow \hat{s}^i_{t+1}$
  Compute reward r_t and termination d_t and add transition $(s_t, a_t, r_t, s_{t+1}, d_t)$ to $\mathcal{D}_{\texttt{model}}$
Return $\mathcal{D}_{\texttt{model}}$

the next H states, starting at state s_t is $p_s(s_{t+1:t+1+H} \mid s_t, a_{t:t+H}, e_{t:t+H}) = \prod_{j=0}^{H} p_s(s_{t+1+j} \mid s_{t+j}, a_{t+j}, e_{t+j})$. Taking the negative log-liklihood yields our training objective for the i-th model:

$$\mathcal{L}(\psi_i) = \frac{1}{HN_{\texttt{e}}} \sum_{t=h}^{N_{\texttt{e}}-H} \sum_{j=0}^{H} \left[\bar{s}^i_{t+1+j} - s_{t+1+j}\right]^{\mathsf{T}} \left(\Sigma^i_{t+j}\right)^{-1} \left[\bar{s}^i_{t+1+j} - s_{t+1+j}\right] + \log\det\Sigma^i_{t+j}\,. \quad (5)$$

Here, $N_{\texttt{e}}$ is the size of a buffer of real-world transitions $\mathcal{D}_{\texttt{env}}$, mean state predictions are propagated deterministically according to $\bar{s}^i_{t+1} = S^i(\bar{s}^i_t, a_t, h_t)$, and each prediction $\bar{s}^i_t$ is used to update the state history h_t. For each state s_t in the buffer, (5) generates a synthetic rollout H steps long and computes a loss related to how much the propagated states differ from the experienced states.

Remark 1. *By optimizing the multi-step loss* (5) *end-to-end, we* i) *force the latent* z_t *to encode information about the environment which is necessary for making state predictions over long horizons and* ii) *average out noisy state estimates and sudden changes in the external torques that occur when the robot makes contact with the ground (Fig. 4). This leads to reliable long-horizon predictions which are substantially more accurate than those generated by black-box models (Fig. 5).*

3.4 Policy Optimization

Finally, we introduce the Semi-Structured Reinforcement Learning (SSRL) in Algorithm 2. This policy optimization strategy is a Dyna-style algorithm [3, 6] which leverages model-predictions as a source of data augmentation. Specifically, this approach branches hallucinated rollouts off real-world trajectories to reduce the effects of model bias, and increases the length of these rollouts throughout training as the model becomes more accurate. We perform many rounds of data generation and policy optimization after collecting real-world samples, and as depicted in Fig. 5 our semi-structured model are able to make substantially more accurate predictions over long horizons when data is scarce. Together, these features enable SSRL to push the sample complexity limits for MBRL. We also found the following necessary for making real-world learning practical:

Deterministic Real-World Rollouts and Random Hallucinations. Steps in the real environment are taken using a deterministic policy μ_θ which simply outputs the mean action from the stochastic policy π_θ. Black-box MBRL approaches typically need to inject random 'dithering' noise to ensure their is enough exploration to obtain accurate models. However, we found that this led to unacceptably erratic behavior in the real-world when paired with our policy class. Recent model-free approaches for learning locomotion in the real-world [32] severely restrict the action space to make stochastic exploration tractable, and thus severely limit the performance of resulting policies. However, our semi-structured models have the right amount of inductive bias to accurately identify the dynamics when the real-world data set $\mathcal{D}_{\texttt{env}}$ is generated with deterministic exploration and data is scarce. We do, however, use a stochastic policy π_θ to generate rich, diverse synthetic data sets $\mathcal{D}_{\texttt{model}}$ for policy optimization. Specifically, policy parameters θ are trained with soft-actor critic (SAC) [33], using a mixture of transitions from real-world and hallucination buffers $\mathcal{D}_{\texttt{env}}$ and $\mathcal{D}_{\texttt{model}}$.

Algorithm 1 Auto-Regressive State Predictions

Inputs hallucination buffer $\mathcal{D}_{\texttt{model}}$, models $\{\hat{p}^i_{\psi_i}\}$, policy π_θ, start state s_0, start history h_0
for $t = 0 \dots k-1$ **do**
  Sample action $a_t \sim \pi_\theta(\cdot \mid s_t, h_t)$
  Randomly choose model $i \sim \mathcal{U}[1, \dots, P]$ and predict next state $\hat{s}^i_{t+1}$ with (4)
  Update state history h_{t+1} with $\hat{s}^i_{t+1}$ and $s_{t+1} \leftarrow \hat{s}^i_{t+1}$
  Compute reward r_t and termination d_t and add transition $(s_t, a_t, r_t, s_{t+1}, d_t)$ to $\mathcal{D}_{\texttt{model}}$
Return $\mathcal{D}_{\texttt{model}}$

从状态 s_t 开始的下一个 H 状态为 $p_s(s_{t+1:t+1+H} \mid s_t, a_{t:t+H}, e_{t:t+H}) = \prod_{j=0}^{H} p_s(s_{t+1+j} \mid s_{t+j}\ a_{t+j}\ e_{t+j})$，，。取负对数似然得到我们针对第 i 个模型的训练目标：

$$\mathcal{L}(\psi_i) = \frac{1}{HN_{\texttt{e}}} \sum_{t=h}^{N_{\texttt{e}}-H} \sum_{j=0}^{H} \left[\bar{s}^i_{t+1+j} - s_{t+1+j}\right]^{\mathsf{T}} \left(\Sigma^i_{t+j}\right)^{-1} \left[\bar{s}^i_{t+1+j} - s_{t+1+j}\right] + \log\det\Sigma^i_{t+j}\,. \quad (5)$$

在此，$N_{\texttt{e}}$ 是真实世界转移的缓冲区大小 $\mathcal{D}_{\texttt{env}}$，平均状态预测根据 $\bar{s}^i_{t+1} = S^i(\bar{s}^i_t, a_t, h_t)$进行确定性传播，且每个预测 $\bar{s}^i_t$ 用于更新状态历史 h_t。对于缓冲区中的每个状态 s_t，(5) 生成一个合成 rollout H 步长，并计算一个损失，该损失反映传播状态与经历状态之间的差异程度。

注 1.通过端到端优化多步损失 (5)，我们 i)迫使隐变量 z_t 编码关于环境的信息，这些信息对于在长视野下进行状态预测是必要的，并且 ii) 能够平均掉噪声状态估计以及机器人与地面接触时发生的外部力矩突变（图 4）。这导致了可靠的长视野预测，其准确性显著高于黑箱模型生成的预测（图 5）。

3.4 策略优化

最后，我们在SSRL中引入了半结构化强化学习（算法 2）。该策略优化策略是一种 Dyna 风格算法 [3, 6]，它利用模型预测作为数据增强的来源。具体而言，该方法从真实世界轨迹中分支出幻觉轨迹，以减少模型偏差的影响，并随着模型变得更加准确，在训练过程中增加这些轨迹的长度。我们在收集真实世界样本后执行多轮数据生成和策略优化，正如图 5 所示，当数据稀缺时，我们的半结构化模型能够在长 horizon 上做出 substantially 更准确的预测。Together，这些特性使SSRL 能够突破 MBRL 的样本复杂度极限。我们还发现以下措施对于使真实世界学习变得实用是必要的：

确定性真实世界轨迹与随机幻觉。在真实环境中执行的步骤采用确定性策略 μ_θ，该策略仅输出随机策略 π_θ 的平均动作。黑箱 MBRL 方法通常需要注入随机“抖动”噪声，以确保有足够的探索来获取准确的模型。然而，我们发现，当与我们的策略类结合使用时，这会导致真实世界中出现不可接受的 erratic 行为。近期用于在真实世界中学习运动的无模型方法 [32] 严重限制了动作空间，以使随机探索变得可行，从而严重限制了所得策略的性能。然而，我们的半结构化模型具有适量的归纳偏置，能够在真实世界数据集 $\mathcal{D}_{\texttt{env}}$ 通过确定性探索生成且数据稀缺时，准确识别动力学特性。不过，我们确实使用随机策略 π_θ 来生成丰富、多样的合成数据集 $\mathcal{D}_{\texttt{model}}$，以用于策略优化。具体而言，策略参数 θ 使用软演员 - 评论家（SAC） [33], 进行训练，并利用来自真实世界和幻觉缓冲区 $\mathcal{D}_{\texttt{env}}$ 与 $\mathcal{D}_{\texttt{model}}$ 的转移混合数据。

Algorithm 2 `SSRL` : Policy Optimization with Semi-structured Dynamics Models

Initialize models $\hat{p}_{\psi_i}$, policy π_θ, critics Q_{ϕ_i}
for $N_{\texttt{epochs}}$ epochs **do**
  Take N_E steps in the environment deterministically with μ_θ and add transitions to $\mathcal{D}_{\texttt{env}}$
  Train models $\hat{p}_{\psi_i}$ on $\mathcal{D}_{\texttt{env}}$ using loss (5)
  Increase hallucination rollout length k according to predetermined schedule
  for K hallucination updates **do**
    for M model rollouts **do**
      Sample state s_0 uniformly from $\mathcal{D}_{\texttt{env}}$ and hallucinate with Algorithm 1
    Perform G updates of policy π_θ using mixture of $\mathcal{D}_{\texttt{env}}$ and $\mathcal{D}_{\texttt{model}}$ at ratio $r_\mathcal{D}$

4 Experimental Results

4.1 Real-world Results

We demonstrate our approach through two real-world experiments where a Unitree Go1 quadruped is trained from scratch to achieve maximum speed on both hard ground and memory foam.

Experimental setup. Training is performed from scratch in the real world over $N_{\texttt{epochs}} = 18$ epochs with $N_E = 1000$ environment steps per epoch, totaling $18,000$ steps or $3.0\,\text{min}$ of real-world interaction. We use an observation history length of $h = 5$, a multi-step loss horizon of $H = 4$, and hallucinate synthetic rollouts for up to $k = 20$ steps; see Appendix D for hyperparameters. To enhance plasticity, the model, actor, and critic are reset at $10,000$ steps [34]. Joint states are measured via encoders, IMU data provides body orientation and angular velocity, and linear velocity is acquired with a Vicon motion capture system. Neural networks are trained in JAX [35], and low-level joint angle commands are sent via Unitree's ROS interface [36]. We compute the mass matrix, Coriolis terms, and gravity vector in (3) using the differentiable simulator Brax [37], enabling gradient-based training of the semi-structured dynamics model with (5).

Results. Fig. 1 shows a time-lapse of rollouts as training progresses. After just $3.0\,\text{min}$ of real-world data, the quadruped achieves an average velocity of $0.98\,\text{m}\,\text{s}^{-1}$ on hard ground and $0.53\,\text{m}\,\text{s}^{-1}$ on memory foam (Fig. 4, right). Figure 4 (left) plots the reward per episode until the first termination ($d_t = 1$). Initially, the quadruped often falls, resulting in low rewards, but after $1.5\,\text{min}$, the learned policy becomes robust, and rewards increase. Despite the challenge of walking on memory foam, where the robot's feet sink deeply, forward velocity improves consistently in both terrains, demonstrating our approach's versatility to differing contact dynamics. To examine model accuracy, we compare the learned external force predictions $\bar{\tau}_t^{e,i}$ to real-world external force estimates τ^e over one second. We estimate the real-world external force with $\tau^e = M(q_t)\ddot{q}_t + C(q_t, \dot{q}_t) + G(q_t) - B\tau_t$ where joint accelerations are estimated by finite differencing: $\ddot{q}_t = (\dot{q}_{t+1} - \dot{q}_t)/\Delta t$ and motor torques τ_t are estimated with the PD control law. Figure 4 (right) shows the learned external vertical force predicted on the base of the robot. Notably, the predictions appear as smoothed versions of the actual torque estimates. This, when combined with the accuracy of our predictions over long-horizons (Section 4.2) provides insight into why our approach enables such effective policy optimization [38]. Appendix B.1 provides plots for the force estimated on additional degrees of freedom for the robot.

4.2 Simulated Experiments

In addition to the results presented here, we provide extensive ablations on standard RL benchmarks in Appendix C. Here, we investigate the following hypotheses:

Hypothesis 1. *`SSRL` boosts performance and sample efficiency for MBRL in contact-rich settings.*

Hypothesis 2. *Training with a multi-step loss outperforms a single-step loss.*

Hypothesis 3. *Predictions from semi-structured dynamics models demonstrate greater accuracy and improved generalization beyond training data compared to black-box models.*

Algorithm 2 `SSRL` : Policy Optimization with Semi-structured Dynamics Models

Initialize models $\hat{p}_{\psi_i}$, policy π_θ, critics Q_{ϕ_i}
for $N_{\texttt{epochs}}$ epochs **do**
  Take N_E steps in the environment deterministically with μ_θ and add transitions to $\mathcal{D}_{\texttt{env}}$
  Train models $\hat{p}_{\psi_i}$ on $\mathcal{D}_{\texttt{env}}$ using loss (5)
  Increase hallucination rollout length k according to predetermined schedule
  for K hallucination updates **do**
    for M model rollouts **do**
      Sample state s_0 uniformly from $\mathcal{D}_{\texttt{env}}$ and hallucinate with Algorithm 1
    Perform G updates of policy π_θ using mixture of $\mathcal{D}_{\texttt{env}}$ and $\mathcal{D}_{\texttt{model}}$ at ratio $r_\mathcal{D}$

4 实验结果

4.1 真实世界结果

我们通过两项真实世界实验展示了我们的方法，其中 Unitree Go1 四足机器人从零开始训练，以在硬地面和记忆海绵上实现最大速度。

实验设置。训练在真实世界中从零开始进行，共经历 $N_{\texttt{epochs}} = 18$轮次，每轮环境步数为 $N_E = 1000$，总计18,000 步骤或3.0 分钟的真实世界交互。我们使用的观测历史长度为 $h = 5$，多步损失范围为$H = 4$，并合成最多 $k = 20$ 步骤的 rollout；详见附录 D 以获取超参数。为增强可塑性，模型、Actor 和 Critic 在第10,000 步骤处重置 [34]。关节状态通过编码器测量，IMU 数据提供机体姿态和角速度，线速度则通过 Vicon 动作捕捉系统获取。神经网络在 JAX [35],中进行训练，底层关节角度指令通过 Unitree 的 ROS 接口发送 [36]。我们利用可微分模拟器 Brax [37],计算公式（3）中的质量矩阵、科里奥利项和重力向量，从而能够基于公式（5）对半结构化动力学模型进行基于梯度的训练。

结果。图 1 展示了随着训练进展的轨迹延时摄影。仅使用 3.0 分钟的真实世界数据后，该四足机器人在硬地面上实现了 0.98 $\text{m}\,\text{s}^{-1}$的平均速度，在记忆海绵上实现了 0.53 $\text{m}\,\text{s}^{-1}$的平均速度（图 4，右）。图 4（左）绘制了直到首次终止($d_t = 1$)为止的每回合奖励。起初，四足机器人经常摔倒，导致奖励较低，但 1.5 分钟后，学习策略变得鲁棒，奖励随之增加。尽管在记忆海绵上行走具有挑战性（机器人的脚会深深陷入），前向速度在两种地形中均持续改善，证明了我们的方法对不同接触动力学的适应性。为了检验模型精度，我们将学习到的外力预测 $\bar{\tau}^{e,i}_t$ 与一秒内的真实世界外力估计 τ^e 进行了比较。我们利用 $\tau^e = M(q_t)\ddot{q}_t + C(q_t, \dot{q}_t) + G(q_t) - B\tau_t$估计真实世界的外力，其中关节加速度通过有限差分估计：$\ddot{q}_t = (\dot{q}_{t+1} - \dot{q}_t)/\Delta t$，电机扭矩$\tau_t$ 则通过 PD 控制律估计。图 4（右）显示了在机器人基座上预测的学习到的外部垂直力。值得注意的是，这些预测表现为实际扭矩估计的平滑版本。这一点，结合我们在长期上的预测准确性（第 4.2 节），揭示了为何我们的方法能够实现如此有效的策略优化 [38]。附录 B.1 提供了机器人额外自由度上受力估计的图表。

4.2 模拟实验

除了此处展示的结果外，我们还在附录 C中提供了对标准强化学习基准测试的广泛消融研究。在此，我们探讨以下假设：

假设 1。 *`SSRL`* 提升了基于模型的强化学习在丰富接触场景中的性能与样本效率。

假设 2。使用多步损失进行训练优于单步损失。

假设 3。与黑箱模型相比，半结构化动力学模型的预测展现出更高的准确性，并在训练数据之外具有更好的泛化能力。

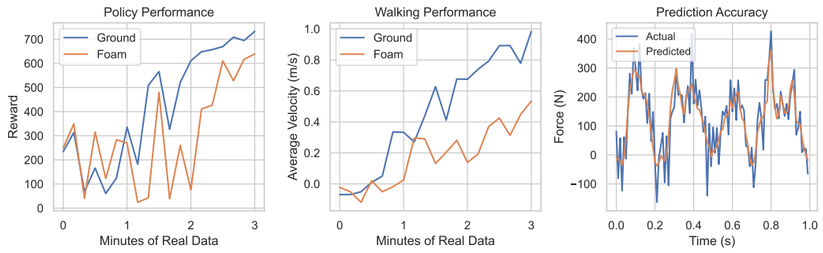


Figure 4: Real-world results. Left—SSRL efficiently performs policy optimization, even when data is scarce. Center—With our approach, the quadruped steadily learns to walk faster. Right—Predicted and real external vertical force acting on the robot base over one second of real-world data. Real forces are estimated by finite differences. The predictions add noticeable smoothing to the real-world data.

Experimental setup. The simulation setup mirrors the real-world setup (Section 4.1), except rollouts are simulated in Brax [37]; parameters are detailed in Appendix D. To examine Hypothesis 1, we perform policy optimization with SSRL (Algorithm 2) and compare its policy performance to a baseline approach that utilizes the same optimization process but substitutes the semi-structured dynamics model with a black-box model. We also benchmark against SAC [33], with an extended run shown in Appendix B.6. To test Hypothesis 2, we repeat the prior experiment, except we train the model with a single-step loss horizon ($H = 1$). To assess generalization (Hypothesis 3), we train our semi-structured models and the black-box models from scratch over 3 minutes of saved simulated data using 1- and 4-step losses. We then generate new data using a stochastic policy where we lower the friction and ground contact stiffness by 25% in the simulator. We generate 20-step synthetic rollouts from 400 randomly-sampled starting states within the new dataset and average the prediction error $\|\hat{s}_t - s_t\|/\dim(s_t)$ over the 400 rollouts. All runs are repeated across 4 random seeds.

Results. Shown in Fig. 5 (left), our semi-structured dynamics models outperform black-box models in both sample efficiency and maximum reward, supporting Hypothesis 1. Even when data is scarce, our semi-structured model generates more accurate synthetic rollouts during exploration, resulting in significantly improved policy performance. We also observe in Fig. 5 (left) that, while black-box models yield similar policy performance for both single- and multi-step losses, training our semi-structured dynamics models with a multi-step loss results in improved performance over a single-step loss, confirming Hypothesis 2. In Fig. 5 (right), we see that our semi-structured models produce predictions 20 steps into the future that are significantly more accurate than black-box models. By leveraging physics-based knowledge, our models generate synthetic rollouts that generalize better to an unseen environment, confirming Hypothesis 3; additional experiments are found in Appendix B.

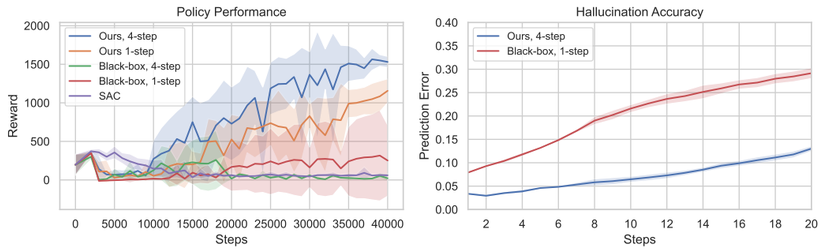


Figure 5: Left—SSRL achieves better policy performance compared to a baseline using black-box models. Right—Prediction error for 20-step synthetic rollouts in an unseen environment showcases our method's superior ability to generalize.

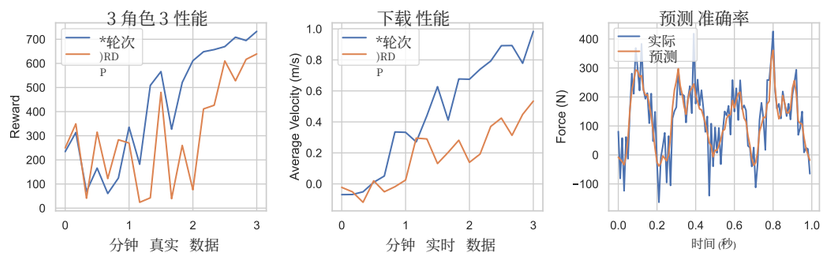


图 4：真实世界结果。左—即使数据稀缺，SSRL 也能高效执行策略优化。中—采用我们的方法，四足机器人稳步学会走得更快。右—基于一秒真实世界数据预测的与真实的外部垂直力作用于机器人基座的情况。真实力通过有限差分估算。预测结果为真实世界数据增添了明显的平滑效果。

实验设置。模拟设置与现实世界设置一致（第 4.1 节），区别在于轨迹是在 Brax [37]中进行模拟的；参数详见附录 D。为了检验假设 1，我们使用SSRL （算法 2）进行策略优化，并将其策略性能与一种基线方法进行比较；该基线方法采用相同的优化流程，但用黑箱模型替代了半结构化动力学模型。我们还以 SAC [33], 为基准进行了对比，其扩展运行结果见附录 B.6。为了测试假设 2，我们重复了前述实验，不同之处在于我们使用单步损失范围（$H = 1$）来训练模型。为了评估泛化能力（假设 3），我们从零开始，利用3 分钟保存的模拟数据，采用1步和4步损失来训练我们的半结构化模型和黑箱模型。随后，我们使用随机策略生成新数据，在该策略中我们将模拟器中的摩擦和地面接触刚度降低了25% 。我们从新数据集中随机采样的400 个起始状态生成了20步合成 rollout，并平均计算了这400 次 rollout 上的预测误差 $\|\hat{s}_t - s_t\|/\dim(s_t)$ 。所有实验均在 4 个随机种子上重复进行。

结果。如图 5（左）所示，我们的半结构化动力学模型在样本效率和最大奖励方面均优于黑盒模型，支持了假设 1。即使数据稀缺，我们的半结构化模型在探索过程中也能生成更准确的合成 rollout，从而显著提升策略性能。我们还在图 5（左）中观察到，虽然黑箱模型在单步损失和多步损失下产生的策略性能相似，但使用多步损失训练我们的半结构化动力学模型相比单步损失能带来性能提升，证实了假设 2。在图 5（右）中，我们可以看到，我们的半结构化模型生成的未来 20 步预测显著比黑箱模型更准确。通过利用基于物理的知识，我们的模型生成的合成 rollout 能够更好地泛化到

未见环境，证实了假设 3；额外实验见附录 B。

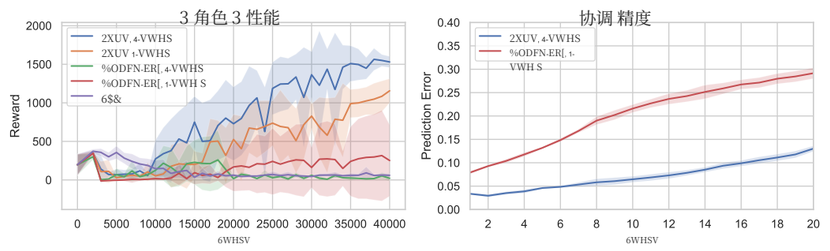


图 5：左—与使用黑盒模型的基线相比，SSRL 实现了更优的策略性能。右—在未见环境中进行 20 步合成 rollout 的预测误差展示了我们方法卓越的泛化能力。

5 Related Work

Model-based RL. Our works builds on a wealth of prior works that use general function approximators and probabilistic modeling for black-box modeling [39, 6, 27]. Model-based reinforcement learning algorithms either learn a model that is used for online planning [40, 27, 41] or Dyna-style algorithms which hallucinate imagined rollouts for direct policy optimization [6, 3, 4]. Moreover, significant attention been devoted to extending these strategies to learn latent representations for environments from available observations [2, 42]. However, these approaches do not leverage known structure to accelerate learning. In contrast, there has been a substantial line of work which leverages known structure to efficiently learn accurate models [15, 16, 17, 19, 15]. However, these approaches generally require access to the full state of the system at inference time, which is impractical for real-world learning for contact-rich with on-board perception. Thus, our semi-structured approach brings the strengths of both paradigms to bear in a single framework. Finally, we also note related work [43, 44, 38] which investigates smoothing out contact dynamics to make policy optimization more tractable for difficult contact-rich problems. We found that our learned models naturally learned smooth representations for the dynamics which generate accurate long-horizon predictions (Fig. 5).

Model-free RL. A parallel line of work [9, 10, 28] aims to make off-policy model-free algorithms (which form the back-bone for our policy optimization strategy) more stable and efficient in low-data regimes. These approaches introduce regularization techniques which enable the use of higher update-to-data ratios without overfitting to the available data, matching the efficiency of model-free methods such as the MBPO [6] algorithm that we build upon. These algorithmic advances are generally orthogonal to our contribution, and thus in the future we plan to incorporate them into our framework to further accelerate real-world learning.

Learning Locomotion Strategies in the Real World. Learning locomotion behaviors from scratch directly in the real-world has primarily been studied in the context of model-free reinforcement learning [45, 46, 32, 47], with a few works using black-box models in the context of model-based reinforcement learning [48, 49]. Compared to these works, our semi-structured modeling approach enable the robot to achieve more dynamic locomotion strategies than these previous approaches, with just a fraction of the real-world samples. Specifically, our approach either achieves a significantly higher walking speed than each of these approaches, or improves on their sample complexity by approximately an order of magnitude (Fig. 4). Several other works investigate fine-tuning locomotion controllers trained in simulation to reduce the burden on real-world data [28, 50] – as we discuss below, we hope to investigate this direction in the near future.

Direct Transfer From Simulation. There has also been recent and rapid progress directly transferring locomotion controllers from simulation zero-shot [51, 26, 25, 52, 53], using techniques such as domain adaptation and domain randomization. In this paper we have focused on learning locomotion controllers from scratch, in an effort to demonstrate the ability of our framework to substantially adapt the behavior of the robot with small amounts of real-world data. However, in the future we plan to fine-tune policies that have been trained using extensive simulated experience, improving the performance of these policies in cases where they fail [28] but leveraging a better initialization for the policy for real-world learning.

6 Limitations

This paper presents a novel framework for model-based reinforcement learning, which leverages physics-informed, semi-structured dynamics models to enable highly sample-efficient policy learning in the real world. However there are several key limitations. First, our method requires observability of enough proprioceptive states to propagate the Lagrangian dynamics of the robot. Additionally, relying solely on proprioception restricts the model's ability to predict changes to the environment such as the appearance of an obstacle or transitions between different ground surfaces. In the future we plan to extend the current framework to include additional perceptual modalities which can infer more about the state of the environment around the robot.

5 相关工作

基于模型的强化学习。我们的工作建立在大量先前研究的基础之上，这些研究利用通用函数逼近器和概率建模进行黑箱建模 [39, 6, 27]。基于模型的强化学习算法要么学习一个用于在线规划的模型 [40, 27, 41]，要么采用 Dyna 风格算法，通过生成想象推演来直接进行策略优化 [6, 3, 4]。此外，大量研究致力于扩展这些策略，以从可用观测中学习环境的潜在表征 [2, 42]。然而，这些方法并未利用已知结构来加速学习。相比之下，已有大量研究工作利用已知结构来高效学习准确模型 [15, 16, 17, 19, 15]。然而，这些方法通常要求在推理时能够访问系统的完整状态，这对于具有丰富接触且依赖机载感知的现实世界学习而言并不切实际。因此，我们的半结构化方法将这两种范式的优势融合到一个统一的框架中。最后，我们还注意到相关研究[43, 44, 38] 探讨了平滑接触动力学，以使策略优化在处理困难的丰富接触问题时更加可行。我们发现，我们学习的模型自然地学到了动力学的平滑表征，从而能够生成准确的长期预测（图 5）。

无模型强化学习。另一条并行的研究路线 [9, 10, 28] 旨在使离策略无模型算法（构成我们策略优化策略的主干）在低数据机制下更加稳定和高效。这些方法引入了正则化技术，使得能够在不过拟合可用数据的情况下使用更高的更新数据比，从而达到了与我们所构建的 MBPO [6]算法等无模型方法相当的效率。这些算法进展通常与我们的贡献正交，因此我们计划在未来将它们纳入我们的框架，以进一步加速真实世界学习。

真实世界中的运动策略学习。从头开始在真实世界中学习运动行为主要是在无模型强化学习的背景下进行研究的 [45, 46, 32, 47], 仅有少数工作在基于模型的强化学习背景下使用了黑箱模型 [48, 49]。与这些工作相比，我们的半结构化建模方法使机器人能够以仅占先前方法一小部分的真实世界样本，实现更具动态性的运动策略。具体而言，我们的方法要么实现了比每种先前方法显著更高的行走速度，要么将它们的样本复杂度提高了约一个数量级（图 4）。其他几项工作研究了微调在模拟中训练的运动控制器，以减轻对真实世界数据的依赖 [28, 50]——正如我们在下文所讨论的，我们希望在不久的将来探索这一方向。

从仿真直接迁移。近期也取得了快速进展，能够利用域适应和域随机化等技术，将运动控制器从模拟环境零样本 [51, 26, 25, 52, 53], 直接迁移。本文专注于从头开始学习运动控制器，旨在证明我们的框架能够利用少量真实世界数据，显著调整机器人的行为。然而，未来我们计划对经过大量模拟经验训练的策略进行微调，在这些策略失效的情况下提升其性能 [28]，同时为真实世界学习提供更好的策略初始化。

6 局限性

本文提出了一种新颖的基于模型的强化学习框架，该框架利用物理信息驱动的半结构化动力学模型，在真实世界中实现高样本效率的策略学习。然而，该方法存在几个关键局限性。首先，我们的方法要求能够观测到足够的本体感受状态，以传播机器人的拉格朗日动力学。此外，仅依赖本体感受限制了模型预测环境变化的能力，例如障碍物的出现或不同地面表面之间的转换。未来，我们计划扩展当前框架，纳入额外的感知模态，从而推断更多关于机器人周围环境状态的信息。

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12

A Implementation Details

In this appendix, we provide details of our implementation for the Unitree Go1 Quadruped, including the observation and action spaces, the reward function, the termination condition, and the control architecture.

A.1 Observation and Action Spaces

The observation space $\Omega \subset \mathbb{R}^{36}$ consists of the elements in Table 1. The x-axis of the base is the forward direction, the y-axis is the leftward direction, and the z-axis is the upward direction. The phase variable $\phi \in [0, 2\pi)$ represents progression along the gait cycle and is defined as $\phi_t = 2\pi t / T_\phi \mod (2\pi)$ where $T_\phi = 0.5\,\mathrm{sec}$ is the gait cycle period.

Observation	Symbol	Dimension
Quaternion orientation of the base	φ	4
Joint angles	q^j	12
Base linear velocity (local frame)	(v^x, v^y, v^z)	3
Base angular velocity (local frame)	$(\omega^x, \omega^y, \omega^z)$	3
Joint speeds	$\dot{q}^j$	12
Cosine of phase	$\cos\phi$	1
Sine of phase	$\sin\phi$	1

Table 1: Observation space.

The action space $\mathcal{A} \subset \mathbb{R}^9$ outputs the change in nominal height for the gait generator and offsets to nominal foot positions from the gait generator, as defined in Table 2.

Action	Symbol	Dimension	Min.	Max.
x-foot position changes	Δp^x	4	$-0.15\,\mathrm{m}$	$0.15\,\mathrm{m}$
y-foot position changes	Δp^y	4	$-0.075\,\mathrm{m}$	$0.075\,\mathrm{m}$
Change in gait generator nominal height	$\Delta h^{\texttt{Gait}}$	1	$-0.1\,\mathrm{m}$	$0.0\,\mathrm{m}$

Table 2: Action space.

A.2 Reward Function and Termination Condition

Reward Function. The reward function is a weighted sum of the terms in Table 3. We set the weights and use exponentials in most of the terms to normalize the reward such that a forward velocity of $1.0\,\mathrm{m\,s^{-1}}$ with maximal values for all other terms will result in a reward of approximately 1.0 for a single time step. The roll φ^x, pitch φ^y, and yaw φ^z of the base are obtained from the base quaternion φ. We define $a \odot b$ as the element-wise multiplication of vectors a and b. Actual torques output from the joint-level PD controllers are not available; we estimate the torque applied at the joint with (11). We define the following `LinearLimit` function which linearly penalizes the torque applied at the j-th joint τ^j when exceeding torque limits; within torque limits, the function is a decaying exponential:

$$\texttt{LinearLimit}(\tau^j, \tau^j_{\texttt{min}}, \tau^j_{\texttt{max}}) = \begin{cases} \tau^j - \tau^j_{\texttt{min}} - 1 & \text{if } \tau < \tau^j_{\texttt{min}} \\ -\exp\left[-\tau^j + \tau^j_{\texttt{min}}\right] & \text{if } \tau^j_{\texttt{min}} \le \tau^j < 0 \\ -\exp\left[\tau^j - \tau^j_{\texttt{max}}\right] & \text{if } 0 \le \tau^j < \tau^j_{\texttt{max}} \\ -\tau^j + \tau^j_{\texttt{max}} - 1 & \text{if } \tau^j \ge \tau^j_{\texttt{max}}. \end{cases} \tag{6}$$

A 实现细节

在本附录中，我们提供了针对 Unitree Go1 四足机器人的实现细节，包括观测空间和动作空间、奖励函数、终止条件以及控制架构。

A.1 观测空间和动作空间

观测空间 $\Omega \subset \mathbb{R}^{36}$ 包含表 1中的元素。 x轴为基座的前进方向， y轴为向左方向， z轴为向上方向。相位变量 $\phi \in [0, 2\pi)$ 表示沿步态周期的进展，定义为 $\phi_t = 2\pi t / T_\phi \mod (2\pi)$ ，其中 $T_\phi = 0.5$秒为步态周期。

观测	符号	维度
基座四元数姿态	φ	4
关节角度	q^j	12
基座线速度（局部坐标系）	$(v^x,\ v^y,\ v^z)$	3
基座角速度（局部坐标系）	$(\omega^x,\ \omega^y,\ \omega^z)$	3
关节速度	$\dot{q}^j$	12
相位余弦	$\cos\ \phi$	1
相位正弦	$\sin\ \phi$	1

表 1：观测空间。

动作空间 $\mathcal{A} \subset \mathbb{R}^9$ 输出步态生成器的标称高度变化量，以及相对于步态生成器生成的标称足端位置的偏移量，如表 2所定义。

动作	符号	维度	Min.	Max.
x 足端位置变化量	Δp^x	4	-0.15 米	0.15 米
y足部位置变化	Δp^y	4	-0.075 米	0.075 米
步态生成器标称高度变化	$\Delta h^{\texttt{Gait}}$	1	-0.1 米	0.0 米

表 2：动作空间。

A.2 奖励函数与终止条件

奖励函数。奖励函数是表 3中各项的加权和。我们设定权重并在大多数项中使用指数，以归一化奖励，使得前向速度为$1.0\,\mathrm{ms^{-1}}$且所有其他项取最大值时，单个时间步的奖励约为 1.0。基座的横滚角 φ^x、俯仰角 φ^y和偏航角 φ^z 由基座四元数 φ获得。我们将 $a \odot b$ 定义为向量 a 和 b的逐元素乘法。关节级 PD 控制器输出的实际扭矩不可用；我们使用公式（11）估算施加在关节上的扭矩。我们定义以下线性限制函数，该函数对超出扭矩限制时施加在第 j个关节 τ^j 上的扭矩进行线性惩罚；在扭矩限制范围内，该函数为衰减指数：

$$\texttt{LinearLimit}(\tau^j, \tau^j_{\texttt{min}}, \tau^j_{\texttt{max}}) = \begin{cases} \tau^j - \tau^j_{\texttt{min}} - 1 & \text{if } \tau < \tau^j_{\texttt{min}} \\ -\exp\left[-\tau^j + \tau^j_{\texttt{min}}\right] & \text{if } \tau^j_{\texttt{min}} \le \tau^j < 0 \\ -\exp\left[\tau^j - \tau^j_{\texttt{max}}\right] & \text{if } 0 \le \tau^j < \tau^j_{\texttt{max}} \\ -\tau^j + \tau^j_{\texttt{max}} - 1 & \text{if } \tau^j \ge \tau^j_{\texttt{max}}. \end{cases} \tag{6}$$

Reward Term	Expression	Weight
Maximize forward velocity	v^x_{t+1}	0.42
Limit base yaw rate	$\exp\left[-(\omega^z_{t+1})^2/0.2\right]$	0.11
Limit base roll	$\exp\left[-(\varphi^x_{t+1})^2/0.25\right]$	0.05
Limit base pitch	$\exp\left[-(\varphi^y_{t+1})^2/0.25\right]$	0.05
Limit base yaw	$\exp\left[-(\varphi^z_{t+1})^2/0.07\right]$	0.11
Limit base side velocity	$\exp\left[-(v^y_{t+1})^2/0.01\right]$	0.11
Limit vertical acceleration	$\exp\left[-(v^z_{t+1}-v^z_t)^2/0.02\right]$	0.03
Limit base roll rate	$\exp\left[-(\varphi^x_{t+1}-\varphi^x_t)^2/0.001\right]$	0.03
Limit base pitch rate	$\exp\left[-(\varphi^y_{t+1}-\varphi^y_t)^2/0.005\right]$	0.03
Limit energy	$\exp\left[-\left\|\dot{q}^j_{t+1}\odot\tau_{t+1}\right\|^2_1/450\right]$	0.05
Penalize excessive torques	$\sum_j \texttt{LinearLimit}(\tau^j_t, \tau^j_{\texttt{min}}, \tau^j_{\texttt{max}})/12$	0.02

Table 3: Reward function terms. The reward at each time step is a weighted sum of these terms.

Termination Condition. The termination flag d_t stops the accumulation of reward after the quadruped falls and is defined by:

$$d_t = \begin{cases} 1 & \text{if } |\varphi^x_t| > \pi/4 \text{ or } |\varphi^y_t| > \pi/4 \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

A.3 Control Architecture

Here, we give detailed specification of the control architecture introduced in Section 2. Referring to the action space definition Table 2, the policy takes in the current observation and a history of observations and outputs offsets to foot positions and a nominal height for the gait generator: $(\Delta p^x_t, \Delta p^y_t, \Delta h^{\texttt{Gait}}_t) \sim \pi_\theta(\cdot \mid s_t, h_t)$. The gait generator $\texttt{Gait} : [0,1) \times \mathbb{R} \to \mathbb{R}^3$ is open-loop and generates for each leg, walking-in-place foot positions for the quadruped by computing vertical foot position offsets from nominal standing foot positions:

$$\texttt{Gait}(\bar{\phi}^l_t; \Delta h^{\texttt{Gait}}_t) = \begin{cases} p^l_{\texttt{stand}} + \left[0, 0, h^{\texttt{Swing}}\left(1 - \cos\left[2\pi \frac{\bar{\phi}^l_t - r^{\texttt{Gait}}}{1 - r^{\texttt{Gait}}}\right]\right) - \Delta h^{\texttt{Gait}}_t\right] & \text{if } \bar{\phi}^l \geq r^{\texttt{Gait}} \\ p^l_{\texttt{stand}} + [0, 0, \Delta h^{\texttt{Gait}}_t] & \text{otherwise} \end{cases} \tag{8}$$

where $p^l_{\texttt{stand}} \in R^3$ is the nominal standing foot position of the l-th leg, expressed in the local base frame, $h^{\texttt{Swing}} = 0.09\,\text{m}$ is the gait peak swing height, and $r^{\texttt{Gait}} = 0.5$ is that fraction of the time feet should remain in contact with the ground. The normalized phase $\bar{\phi}^l \in [0,1)$ specifies the progress of the l-th leg along its gait cycle and is calculated with:

$$\bar{\phi}^l_t = \left(\frac{\phi_t}{2\pi} + 0.5 + b^l\right) \mod 1, \tag{9}$$

where b^l is the phase bias for the l-th leg; we use a value of 0 for the front-right and rear-left legs, and a value of 0.5 for the front-left and rear-right legs. The desired positions of the l-th foot in the local base frame are given by:

$$p_t(\Delta p^{x,l}_t, \Delta p^{y,l}_t, \Delta h^{\texttt{Gait}}_t, \bar{\phi}^l_t) = \left[\Delta p^{x,l}_t, \Delta p^{y,l}_t, 0\right] + \texttt{Gait}(\bar{\phi}^l_t; \Delta h^{\texttt{Gait}}_t), \tag{10}$$

where $\Delta p^{x,l}_t$ and $\Delta p^{y,l}_t$ are the x- and y-foot positions offsets for the l-th foot from the policy. For each foot, the desired foot positions (10) are computed and sent to an inverse kinematics solver to produce desired joint angles $q^{\texttt{des}} \in \mathbb{R}^{12}$. The desired joint angles are sent to the joint level PD controllers, where the desired torque outputs are:

$$\tau_t = K_p(q^{\texttt{des}} - q^j) - K_p \dot{q}^j, \tag{11}$$

and we use proportional gain $K_p = 112\,\text{N}\,\text{m}\,\text{rad}^{-1}$ and derivative gain $K_p = 3.5\,\text{N}\,\text{m}\,\text{s}\,\text{rad}^{-1}$.

奖励项	表达式	权重
最大化前向速度	v^x_{t+1}	0.42
限制基座偏航速率	$\exp\left[-(\omega^z_{t+1})^2/0.2\right]$	0.11
限制基座横滚	$\exp\left[-(\varphi^x_{t+1})^2/0.25\right]$	0.05
限制基座俯仰	$\exp\left[-(\varphi^y_{t+1})^2/0.25\right]$	0.05
限制基座偏航	$\exp\left[-(\varphi^z_{t+1})^2/0.07\right]$	0.11
限制基座侧向速度	$\exp\left[-(v^y_{t+1})^2/0.01\right]$	0.11
限制垂直加速度	$\exp\left[-(v^z_{t+1}-v^z_t)^2/0.02\right]$	0.03
限制基座横滚角速度	$\exp\left[-(\varphi^x_{t+1}-\varphi^x_t)^2/0.001\right]$	0.03
限制基座俯仰角速度	$\exp\left[-(\varphi^y_{t+1}-\varphi^y_t)^2/0.005\right]$	0.03
限制能量	$\exp\left[-\left\|\dot{q}^j_{t+1}\odot\tau_{t+1}\right\|^2_1/450\right]$	0.05
惩罚过大扭矩	$\sum_j$ 线性限制 $(\tau^j_t, \tau^j_{\texttt{min}}, \tau^j_{\texttt{max}})/12$	0.02

表 3：奖励函数项。每个时间步的奖励是这些项的加权和。

终止条件。终止标志 d_t 在四足机器人摔倒后停止奖励累积，其定义如下：

$$d_t = \begin{cases} 1 & \text{if } |\varphi^x_t| > \pi/4 \text{ or } |\varphi^y_t| > \pi/4 \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

A.3 控制架构

在此，我们详细介绍了第2 节中引入的控制架构。参考动作空间定义表 2，策略接收当前观测和历史观测，并输出足端位置偏移和用于步态生成器的标称高度：$(\Delta p^x_t, \Delta p^y_t, \Delta h^{\texttt{Gait}}_t) \sim \pi_\theta(\cdot \mid s_t, h_t)$。步态生成器 $\texttt{Gait} : [0,1) \times \mathbb{R} \to \mathbb{R}^3$ 为开环系统，通过计算相对于名义站立足端位置的垂直足端位置偏移，为四足机器人的每条腿生成原地行走的足端位置：

$$\texttt{Gait}(\bar{\phi}^l_t; \Delta h^{\texttt{Gait}}_t) = \begin{cases} p^l_{\texttt{stand}} + \left[0, 0, h^{\texttt{Swing}}\left(1 - \cos\left[2\pi \frac{\bar{\phi}^l_t - r^{\texttt{Gait}}}{1 - r^{\texttt{Gait}}}\right]\right) - \Delta h^{\texttt{Gait}}_t\right] & \text{if } \bar{\phi}^l \geq r^{\texttt{Gait}} \\ p^l_{\texttt{stand}} + [0, 0, \Delta h^{\texttt{Gait}}_t] & \text{otherwise} \end{cases}$$

( ) 其中 $p^l_{\texttt{stand}} \in R^3$ 是第 l条腿在局部基座坐标系中的名义站立足端位置， 8

$h^{\texttt{Swing}} = 0.09\text{m}$ 是步态峰值摆动高度，而 $r^{\texttt{Gait}} = 0.5$ 是足部应保持与地面接触的时间比例。归一化相位 $\bar{\phi}^l \in [0,1)$ 指定了第 l条腿在其步态周期中的进度，其计算方式为：

$$\bar{\phi}^l_t = \left(\frac{\phi_t}{2\pi} + 0.5 + b^l\right) \mod 1, \tag{9}$$

其中 b^l 是第 l 条腿的相位偏置；我们对右前腿和左后腿使用值 0，对左前腿和右后腿使用值 0.5。第 l 个足端在局部基座坐标系中的期望位置由下式给出：

$$p_t(\Delta p^{x,l}_t, \Delta p^{y,l}_t, \Delta h^{\texttt{Gait}}_t, \bar{\phi}^l_t) = \left[\Delta p^{x,l}_t, \Delta p^{y,l}_t, 0\right] + \texttt{Gait}(\bar{\phi}^l_t; \Delta h^{\texttt{Gait}}_t), \tag{10}$$

其中 $\Delta p^{x\,l}_t$、和 $\Delta p^{y\,l}_t$、是来自策略的第 x 和第 y 个足端的位置偏移量。l。对于每个足端，计算期望足端位置（10）并将其发送给逆运动学求解器，以生成期望关节角度 $q^{\texttt{des}} \in \mathbb{R}^{12}$。随后将这些期望关节角度发送给关节级 PD 控制器，从而得到期望扭矩输出：

$$\tau_t = K_p(q^{\texttt{des}} - q^j) - K_p \dot{q}^j, \tag{11}$$

我们使用比例增益 $K_p = 112\,\text{N}\,\text{m}\,\text{rad}^{-1}$ 和微分增益 $K_p = 3.5\,\text{N}\,\text{m}\,\text{s}\,\text{rad}^{-1}$。

B Additional Experiments

In this appendix, we present the results of additional experiments which demonstrate our model's ability to make accurate predictions far into the future, generalized to unseen training data, and enable efficient training.

B.1 External Torque Estimate Validation

Contact forces are discontinuous in nature and can be difficult to learn. To this end, we validate the learned external torque predictions $\tau_t^{e,i}$ by comparing them to estimates of the real-world external torques τ^e over one second of real-world data, presented in Fig. 6. Estimates of the real-world external torque are computed as $\tau^e = M(q_t)\ddot{q}_t + C(q_t, \dot{q}_t) + G(q_t) - B\tau_t$ where joint accelerations are estimated by finite differencing: $\ddot{q}_t = (\dot{q}_{t+1} - \dot{q}_t)/\Delta t$ and low-level motor torques τ_t are estimated with the PD control law. The external forces acting on the floating base and the external torques acting on the joints of the front-right leg are shown. Our learned external torque predictions closely align with the estimated actual external torques. Notably, the predictions appear as smoothed versions of the actual torque estimates. Finite differencing of the velocity measurement introduces noise into the actual torque estimates while our multi-step loss learning process inherently smooths the predictions.

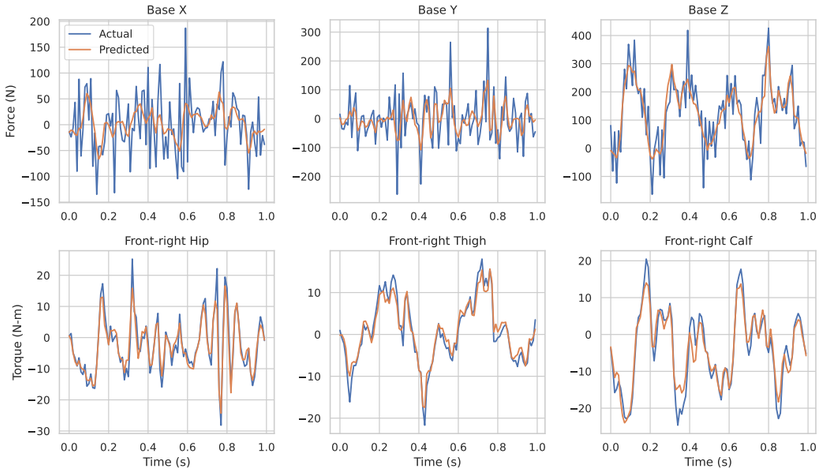


Figure 6: Predicted external forces and actual external force estimates over one second of real-world data for the floating base and the joints of the front-right leg. Our learned external torque predictions closely align with the estimated actual external torques.

B.2 Model Rollout Accuracy

Here, we examine the accuracy of trajectories generated from learned models and the ability of the model to generalize beyond the available training data. First, we train our semi-structured models and the black-box models from scratch over 3 minutes of saved simulated data using 1- and 4-step losses. Using the trained models, 20-step synthetic rollouts are generated from 400 randomly-sampled starting states within the data. The average prediction error $\|\hat{s}_t - s_t\|/\dim(s_t)$ at each time step t is averaged over the 400 trajectories. This is repeated for 4 random seeds where the best results from the 1- or 4-step losses for each model type are recorded. We then repeat this experiment for the real-world dataset. Finally, the experiment is repeated again, except we generate

B 额外实验

在本附录中，我们展示了额外实验的结果，这些结果证明了我们的模型能够对未来进行准确预测、泛化到未见过的训练数据，并实现高效训练。

B.1 外部力矩估计验证

接触力本质上是不连续的，可能难以学习。为此，我们通过将学习到的外部力矩预测，$_t{}^i$ 与真实世界数据中一秒内的实际外部力矩估计值进行比较来验证它们 τ^e，结果见图 6。真实世界外部力矩的估计值计算为 $\tau^e = M(q_t)\ddot{q}_t + C(q_t, \dot{q}_t) + G(q_t) - B\tau_t$，其中关节加速度通过有限差分法估算：$\ddot{q}_t = (\dot{q}_{t+1} - \dot{q}_t)/\Delta t$ 而底层电机力矩 τ_t 则利用 PD 控制律进行估算。图中展示了作用于浮动基座的外力以及作用于右前腿关节的外部力矩。我们学习到的外部力矩预测与实际外部力矩估计值高度吻合。值得注意的是，这些预测表现为实际扭矩估计值的平滑版本。对速度测量进行有限差分会向实际扭矩估计引入噪声，而我们的多步损失学习过程则天然地对预测进行了平滑处理。

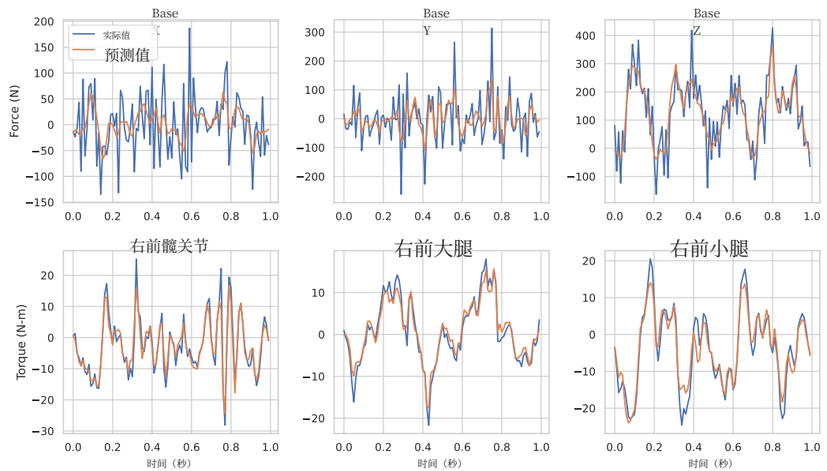


图 6：基于真实世界数据，在一秒时间内预测的外力与浮动基座及右前腿关节的实际外力估计值。我们学习到的外部力矩预测值与实际估计的外部力矩高度吻合。

B.2 模型 rollout 精度

在此，我们检验从学习到的模型生成的轨迹的准确性，以及模型泛化到可用训练数据之外的能力。首先，我们使用 1 步和 4 步损失，基于 3 分钟保存的模拟数据，从头训练我们的半结构化模型和黑盒模型。利用训练好的模型，从数据内随机采样的 400 个起始状态生成 20 步合成 rollout。每个时间步 t 的平均预测误差 $\|\hat{s}_t - s_t\|/\dim(s_t)$ 是在这 400 条轨迹上取平均得到的。该过程针对 4 个随机种子重复进行，并记录每种模型类型在 1 步或 4 步损失下的最佳结果。随后，我们在真实世界数据集上重复此实验。最后，再次重复该实验，只不过我们生成

a new simulated dataset using a stochastic policy and on altered terrain: friction and ground contact stiffness are lowered by 25%. For this final case, the saved models from the first case are evaluated on this new dataset. The results for these 3 experiments are presented in Fig. 7. For all cases, our semi-structured models produce predictions 20 steps into the future that are significantly more accurate than black-box models. The results from the last experiment demonstrate our model's superior ability to generalize to an unseen environment where noise is added to actions.

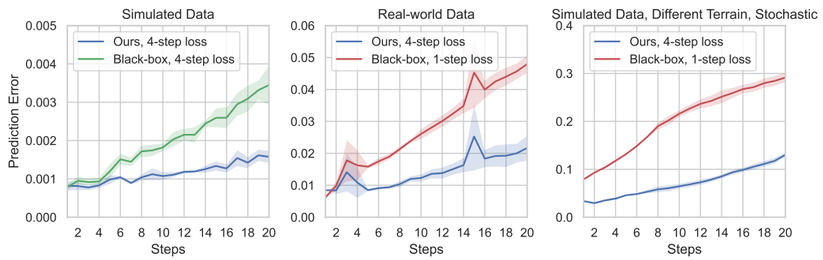


Figure 7: Prediction error for 20-step synthetic rollouts using our semi-structured dynamics models and the black-box models where the best results from the 1- or 4- step losses are presented. Prediction error is averaged over 400 trajectories. Plots show the mean and standard deviation over 4 random seeds.

B.3 Robustness to Errors in the Lagrangian Dynamics

In this experiment, we demonstrate that SSRL performance is robust against errors in *a priori* knowledge of the robot's inertial properties which are used to construct the Lagrangian dynamics (1). To simulate these modeling errors, we randomly vary each link's mass by $\pm 25\%$ and each joint's damping by $\pm 50\%$ for the Go1 environment used for simulated data collection. The remainder of the setup is per Section 4.2 and runs are repeated across 4 random seeds. The results are presented in Fig. 8; even when there are errors in the *a priori* knowledge of the robot's inertial properties, policy performance is similar to runs with no errors. Our external torque estimators (Section 3.1) learn to predict forces to compensate for these errors, resulting in the similar performance.

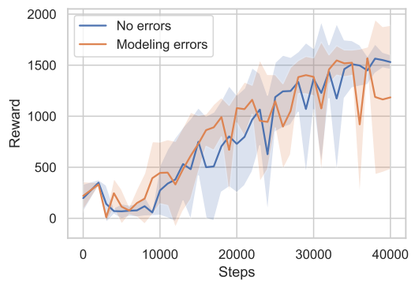


Figure 8: Our approach is robust to errors in *a priori* knowledge of the robot's inertial properties.

B.4 Modeling Uncertainty

Here, we examine the benefit of using the noise estimator and an ensemble of models. The noise estimator prevents overfitting to noise and using an ensemble of models captures epistemic uncertainty

一个使用随机策略并在改变地形上的新模拟数据集：摩擦和地面接触刚度降低了25%。对于这最后一种情况，将在第一种情况下保存的模型在这个新数据集上进行评估。这三项实验的结果展示在图 7中。在所有情况下，我们的半结构化模型产生的未来20 步预测显著比黑盒模型更准确。最后一项实验的结果证明了我们的模型在动作加入噪声的未见环境中具有卓越的泛化能力。

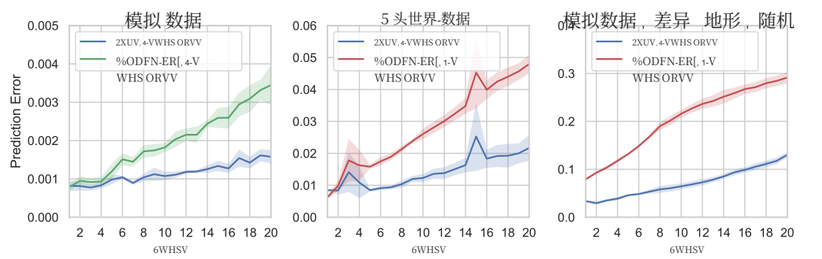


图 7：使用我们的半结构化动力学模型和黑箱模型进行 20 步合成 rollout 的预测误差，展示了来自 1 步或 4 步损失的最佳结果。预测误差取 400 条轨迹的平均值。图表显示了 4 个随机种子上的均值和标准差。

B.3 对拉格朗日动力学中误差的鲁棒性

在本实验中，我们证明了 SSRL 的性能对于机器人惯性特性的 先验 知识误差具有鲁棒性，而这些惯性特性被用于构建拉格朗日动力学（1）。为了模拟这些建模误差，我们在用于模拟数据收集的 Go1 环境中，随机改变每个连杆的质量达 $\pm 25\%$，每个关节的阻尼达 $\pm 50\%$。其余设置遵循 第 4.2 节，并在 4 个随机种子上重复运行。结果展示于 图 8；即使存在机器人惯性特性的 先验 知识误差，策略性能也与无误差运行相似。我们的外部扭矩估计器（第 3.1 节）学会了预测力以补偿这些误差，从而实现了相似的性能。

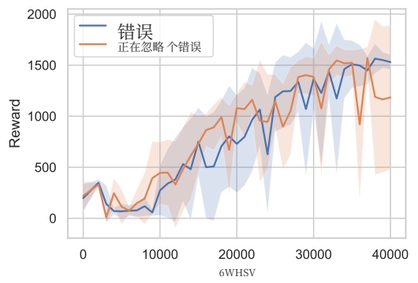


图 8：我们的方法对机器人惯性特性的先验 知识误差具有鲁棒性。

B.4 建模不确定性

在此，我们考察使用噪声估计器和模型集成的益处。噪声估计器可防止对噪声过拟合，而使用模型集成则能捕捉认知不确定性。

[27]. We plot the training performance over 4 random seeds for 3 test cases: (i) using the probabilistic ensemble described in Section 3.1, (ii) removing the noise estimators, and (iii) removing the noise estimators and the ensemble. The results are presented in Fig. 9; the highest performance is obtained with our probabilistic ensemble.

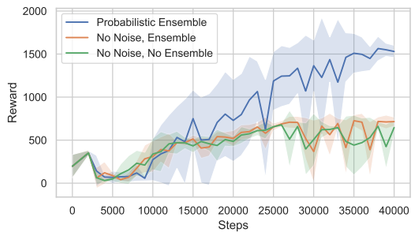


Figure 9: Training performance when removing the noise estimators and removing both the noise estimators and ensemble.

B.5 Additional Simulated Terrain Experiments

To further demonstrate the versatility of our approach on varying contact surfaces, we perform additional experiments in simulation. Within the simulator, we vary the friction coefficient and contact time constant which determines the stiffness of contact. We perform additional training runs for friction coefficients of 0.3 and 0.8, and contact time constants of 0.06 and 0.12; the friction and time constant used for all other simulated experiments in this paper are 0.6 and 0.02 respectively. As depicted in Fig. 10, we find that our method still yields favorable performance even under varying simulated contact conditions.

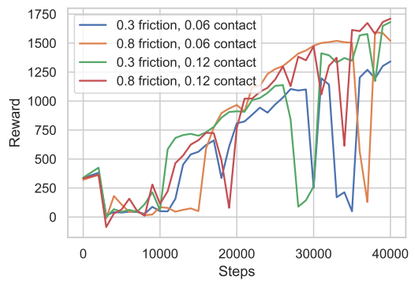


Figure 10: Training performance of our method with varying simulated contact conditions.

B.6 SAC Performance

State-of-the-art real-world quadrupedal locomotion results where policies are trained from scratch [32] utilized SAC-like agents which directly take actions in the environment. However, highly restricted action spaces were required to obtain stable training behavior with 20 minutes of training data. In contrast, we use action spaces which are more in line with standard quadruped results [25, 29, 30]. This enables faster and more dynamic gaits, but also leads to a challenging optimization problem for SAC. Here, we repeat the experiment of Section 4.2, but we allow the SAC training to run for many more samples and present the results in Fig. 11. Even though SAC now converges, it takes 250 times more interaction with the environment than our method.

[27]。我们绘制了在 4 个随机种子下的训练性能，涵盖以下 3 种测试情况：(i) 使用第 3.1 节所述的概率集成，(ii) 移除噪声估计器，以及 (iii) 同时移除噪声估计器和集成。结果如图 9 所示；我们的概率集成取得了最高性能。

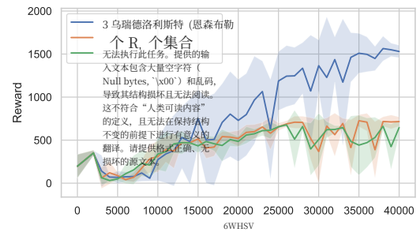


图 9：移除噪声估计器以及同时移除噪声估计器和集成时的训练性能。

B.5 额外模拟地形实验

为了进一步展示我们的方法在不同接触表面上的通用性，我们在模拟器中进行了额外的模拟实验。在模拟器中，我们改变了摩擦系数和决定接触刚度的接触时间常数。我们针对摩擦系数为0.3 和0.8，以及接触时间常数为0.06 和0.12的情况进行了额外的训练运行；本文中所有其他模拟实验所使用的摩擦系数和时间常数分别为0.6 和0.02 。如图 10所示，我们发现即使在不同的模拟接触条件下，我们的方法仍能取得良好的性能。

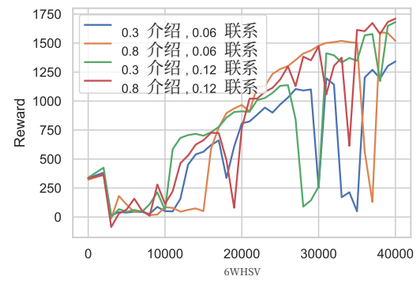


图 10：我们的方法在不同模拟接触条件下的训练性能。

B.6 SAC 性能

从零开始训练策略所取得的业界领先的四足机器人真实世界运动成果[32] 采用了类 SAC 智能体，直接在环境中执行动作。然而，为了在仅使用 20 分钟训练数据的情况下获得稳定的训练行为，必须采用高度受限的动作空间。相比之下，我们使用的动作空间更符合标准的四足机器人研究成果[25, 29, 30]。这使得能够实现更快、更动态的步态，但也给 SAC 带来了极具挑战性的优化问题。在此，我们重复了第 4.2 节中的实验，但允许 SAC 训练运行更多样本，并将结果展示于图 11中。尽管 SAC 如今能够收敛，但其与环境交互的次数仍是我们方法的250 倍。

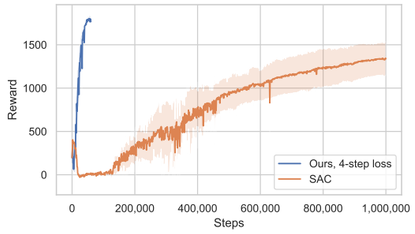


Figure 11: Training curves when running SAC to convergence; SAC requires 250 times more interaction with the environment than our method.

C Simulated Benchmark Experiments

To demonstrate the versatility of our approach, we perform additional simulated experiments using standard, contact-rich, benchmark environments [54] commonly used to evaluate reinforcement learning (RL) algorithms.

Experimental setup. We use the standard MuJoCo [54] environments Hopper, Walker2d, and Ant, which have been implemented as part of Brax [37]. Similar to the quadruped, each of these environments feature a floating-base robot with articulated limbs which make and break contact with the ground to produce motion. However, unlike the Go1 environment, these environments lack structured controllers. Instead, the outputs from the policy are only scaled linearly before being directly applied as torques on the joints. To test Hypothesis 1 and Hypothesis 2, we compare our semi-structured approach trained with a multi-step loss ($H = 4$) to the black-box approach from Section 4.2 trained with the single-step loss ($H = 1$). In both of these cases, the agent acts deterministically within the environment per Algorithm 2. We also benchmark against SAC [33], allowing the agent to act stochastically in the environment for this algorithm only. The hyperparameters used for training are found in Appendix D and all runs are repeated for 4 random seeds.

Results. The results of these experiments are presented in Fig. 12. We observe a significant performance improvement when utilizing our semi-structured models trained with a multi-step loss, compared to the black-box approach trained with a single-step loss, confirming Hypothesis 1 and Hypothesis 2. These results demonstrate that our approach works not only with the Go1 environment, but also with other contact-rich environments with unstructured controllers.

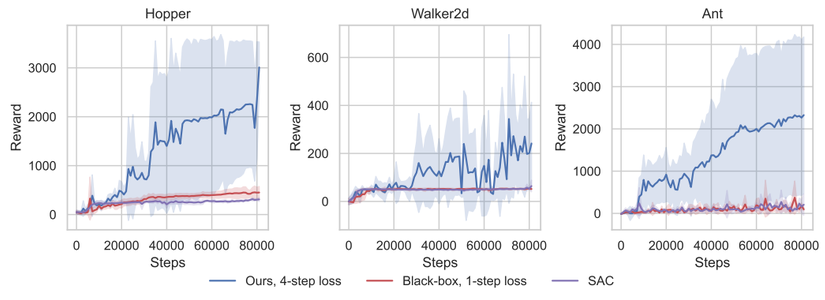


Figure 12: Simulated benchmark results. Better performance is achieved when using our semi-structured dynamics models and a multi-step loss. Plots show the mean and standard deviation for episodic rewards.

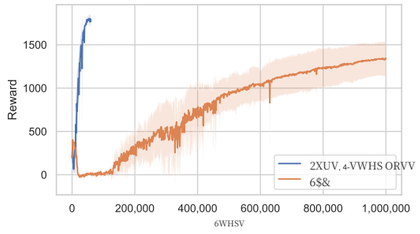


图 11：运行 SAC 至收敛时的训练曲线；SAC 所需的与环境交互次数是我们方法的 250 倍。

C 模拟基准实验

为展示我们方法的通用性，我们在使用标准、丰富接触的基准环境方面开展了额外的模拟实验 [54]，这些环境通常用于评估强化学习 (RL) 算法。

实验设置。我们使用标准的 MuJoCo [54] 环境 Hopper、Walker2d 和 Ant，这些环境已作为 Brax 的一部分实现 [37]。与四足机器人类似，这些环境均包含一个带有铰接肢体的浮动基座机器人，通过与地面接触和分离来产生运动。然而，与 Go1 环境不同，这些环境缺乏结构化控制器。相反，策略的输出仅经过线性缩放后，便直接作为扭矩施加到关节上。为了测试假设 1和假设 2，我们将采用多步损失训练的半结构化方法 ($H = 4$)与第 4.2 节中采用单步损失训练的黑盒方法Section 4.2 （$H = 1$）进行比较。在这两种情况下，智能体均按照算法 2在环境中确定性地进行动作。我们还以 SAC [33], 为基准，仅针对该算法允许智能体在环境中随机地行动。训练所用的超参数见附录 D，所有实验均重复运行 4 次随机种子。

结果。这些实验的结果展示在图 12中。我们观察到，与采用单步损失训练的黑盒方法相比，利用采用多步损失训练的半结构化模型时，性能显著提升，这证实了假设 1和假设 2。这些结果表明，我们的方法不仅适用于 Go1 环境，也适用于其他具有丰富接触和非结构化控制器的环境。

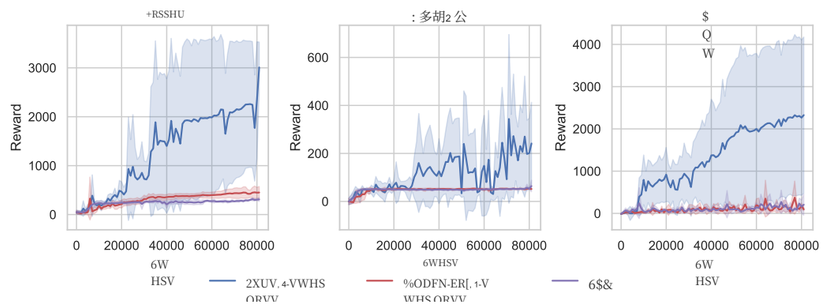


图 12：模拟基准测试结果。使用我们的半结构化动力学模型和多步损失可实现更优性能。图表展示了回合奖励的均值和标准差。

D Experiment Hyperparameters

Table 4 contains the hyperparameters used with our approach; these hyperparameters were also used with the approach that incorporated black-box models. Table 5 contains the SAC hyperparameters used for our approach, the black-box approach, and standard SAC.

Hyperparameter	Go1 (real world)	Go1 (simulated)	Benchmarks
Epochs, $N_{\texttt{epochs}}$	18	40	80
Environment steps per epoch, N_E	1000		
Hallucination updates per epoch, K	$10 \rightarrow 1,000$ over epochs $0 \rightarrow 4$		
Model rollouts per hallucination update, M	400		
Synthetic rollout length, k	$1 \rightarrow 20$ over epochs $0 \rightarrow 10$		$1 \rightarrow 45$ over epochs $0 \rightarrow 15$
Real to synthetic data ratio, $r_{\mathcal{D}}$	0.06		
Gradient updates per hallucination update, G	40	60	20
State history length, h	5		1
Multi-step loss horizon, H	4	1 or 4	
Model learning rate	1×10^{-3}		
Model training batch size	200		

Table 4: Hyperparameters for our approach and the baseline approach with black-box models. $x \rightarrow y$ over epochs $a \rightarrow b$ denotes a clipped linear function, i.e. at epoch i, $f(i) = \texttt{clip}(x + \frac{i-a}{b-a}(y - x), x, y)$.

Hyperparameter	Go1 (real world)	Go1 (simulated)	Benchmarks
Learning rate	2×10^{-3}		3×10^{-3}
Discount factor, γ	0.99		
Batch size	256		
Target smoothing coefficient, τ	1×10^{-3}		5×10^{-3}
Actor network (MLP) width $\times$ depth	512×2		256×2
Critic network (MLP) width $\times$ depth	512×2		256×2

Table 5: SAC hyperparameters used for our approach, the black-box approach, and standard SAC.

D 实验超参数

表 4包含了我们方法所使用的超参数；这些超参数也被用于结合了黑箱模型的方法中。表 5包含了用于我们方法、黑盒方法以及标准 SAC 的 SAC 超参数。

超参数	Go1（真实世界）	Go1（模拟）	基准测试
轮次， $N_{\texttt{epochs}}$	18	40	80
每轮环境步数， N_E	1000		
每轮幻觉更新， K	$10 \rightarrow 1,$ 000（共 $0 \rightarrow 4$轮）		
每次幻觉更新模型 rollout， M	400		
合成 rollout 长度， k	$1 \rightarrow 20$ 跨轮次 $0 \rightarrow 10$		$1 \rightarrow 45$ 每轮次 $0 \rightarrow 15$
真实数据与合成数据比率， $r_{\mathcal{D}}$	0.06		
每次幻觉更新的梯度更新次数， G	40	60	20
状态历史长度， h	5		1
多步损失范围， H	4	1或 4	
模型学习率	1×10^{-3}		
模型训练批量大小	200		

表 4：我们的方法与基于黑盒模型的基线方法的超参数。 $x \rightarrow y$ $a \rightarrow b$ i $f(i) = \texttt{clip}(x + \frac{i-a}{b-a}(y -$ over epochs 表示一个截断线性函数，即在第 个轮次时，x) $x, y)$。

超参数	Go1（真实世界）	Go1（模拟）	基准测试
学习率	2×10^{-3}		3×10^{-3}
折扣因子， γ	0.99		
批量大小	256		
目标平滑系数， τ	1×10^{-3}		5×10^{-3}
Actor 网络（多层感知机）宽度 ×深度	512×2		256×2
Critic 网络（多层感知机）宽度 × 深度	512×2		256×2

表 5：用于我们的方法、黑盒方法和标准 SAC 的 SAC 超参数。